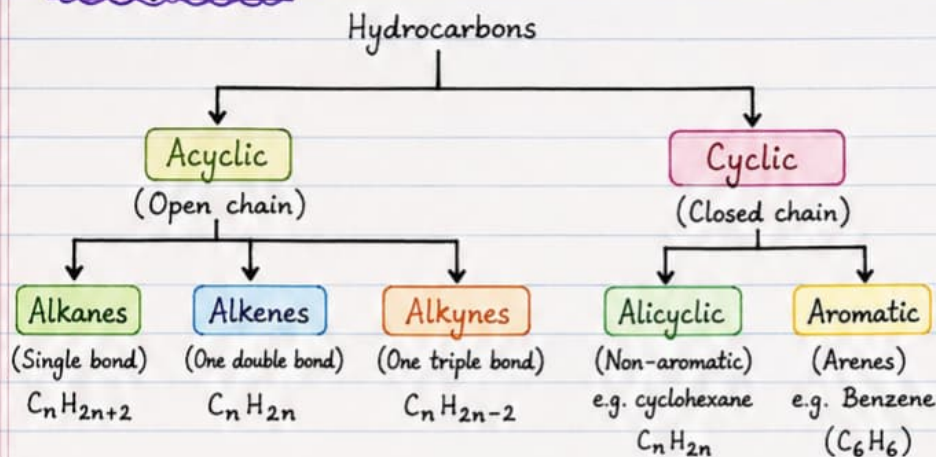


- ★ Hydrocarbons are organic compounds containing **only carbon and hydrogen** atoms. They are the **simplest organic compounds** and the **foundation** of organic chemistry.

Classification

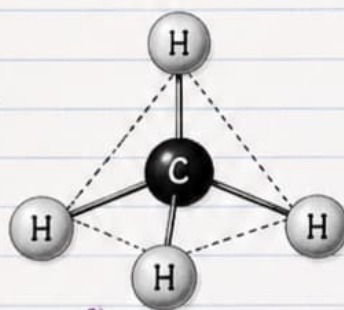


General Formulae

- Alkanes $\rightarrow C_nH_{2n+2}$
 Alkenes $\rightarrow C_nH_{2n}$
 Alkynes $\rightarrow C_nH_{2n-2}$
 Cycloalkanes $\rightarrow C_nH_{2n}$
 Arenes $\rightarrow C_nH_{2n-6}$

1. Alkanes (Saturated Hydrocarbons)

- Contain only **single bond** between carbon atoms.
- General formula $\rightarrow C_nH_{2n+2}$
- Hybridisation $\rightarrow sp^3$
- Bond angle $\rightarrow 109.5^\circ$

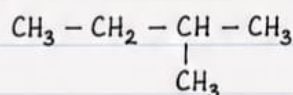


Tetrahedral
structure
(109.5°)

Nomenclature (IUPAC)

- Identify the longest carbon chain.
- Replace '**ane**' suffix for alkanes.
- Number the chain from the end nearer to substituent.
- Write the name of substituent in alphabetical order.

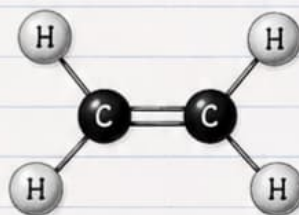
Example :



2-methylbutane

2. Alkenes (Unsaturated Hydrocarbons)

- Contain at least one **C = C** double bond.
- General formula $\rightarrow C_nH_{2n}$
- Hybridisation $\rightarrow sp^2$
- Bond angle $\rightarrow 120^\circ$



Planar
structure
(120°)

Nomenclature (IUPAC)

- Replace '**ane**' with '**ene**'.
- Number the chain to give the double bond the lowest number.
- Indicate the position of double bond.

Example :



But-1-ene



Quick Facts

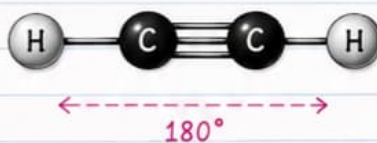
- Alkanes $\rightarrow$ Least reactive
- Alkenes $\rightarrow$ More reactive than alkanes
- Alkynes $\rightarrow$ More reactive than alkenes
- Aromatic $\rightarrow$ Special stability due to resonance

Remember

- ★ More the pi (π) bonds, more will be the reactivity.
- ★ Stable > Less reactive

3. Alkynes (Unsaturated Hydrocarbons)

- Contain at least one $C \equiv C$ triple bond.
- General formula $\rightarrow C_nH_{2n+2}$
- Hybridisation $\rightarrow sp$
- Bond angle $\rightarrow 180^\circ$
- Less reactive than alkenes.



Linear
structure
(180°)

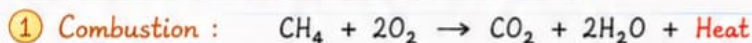
Nomenclature (IUPAC)

- Replace 'ane' with 'yne'.
- Number the chain to give the triple bond the lowest number.
- Indicate the position of triple bond.

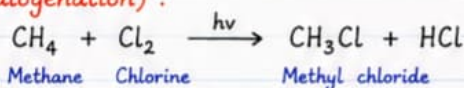
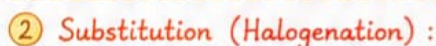
Example :



4. Reactions of Alkanes (Saturated)



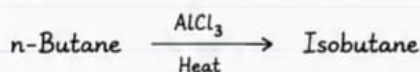
Complete combustion
produces CO_2 and H_2O .



Takes place in presence
of sunlight or UV light.

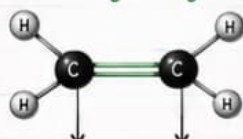
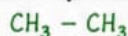
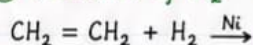


Breaking of large molecules
into smaller ones.



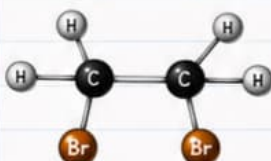
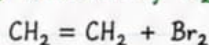
Change in structure without
change in molecular formula.

5. Reactions of Alkenes (Unsaturated)

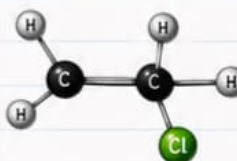
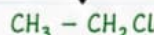
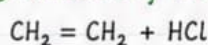


Ethane Ethane

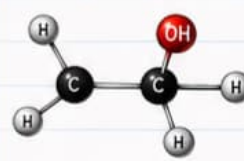
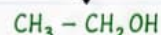
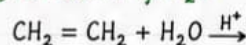
Hydrogenation
(Catalyst: Ni/Pt/Pd)



Vicinal dihalide
(e.g., 1,2-dibromoethane)



Haloalkane
(e.g., chloroethane)



Alcohol
(e.g., ethanol)

☆ All these reactions are electrophilic addition reactions.

6. Comparison of Alkanes, Alkenes & Alkynes

Property	Alkanes	Alkenes	Alkynes
General formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
Bond type	$C - C$	$C = C$	$C \equiv C$
Hybridisation	sp^3	sp^2	sp
Bond angle	109.5°	120°	180°
Shape	Tetrahedral	Trigonal planar	Linear
Reactivity	Least	More	More (than alkenes)



Quick Revision

- ☆ More the number of C-H bonds, more is the stability.
- ☆ Reactivity order:
Alkynes > Alkenes > Alkanes
- ☆ Unsaturated hydrocarbons show addition reactions.
- ☆ Saturated hydrocarbons show substitution reactions.



7. Cyclic Hydrocarbons

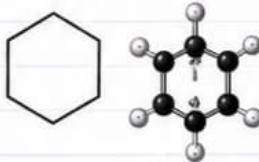
- Hydrocarbons whose carbon atoms are linked to form a ring (closed chain).
- Can be **Alicyclic** (non-aromatic) or **Aromatic**.
- General formula (monocyclic): C_nH_{2n} ($n \geq 3$)

Alicyclic Hydrocarbons

- Contain rings but do not show aromatic character.
- Examples: Cyclopropane, Cyclobutane, Cyclohexane.

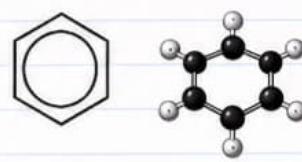
Alicyclic

(Non-aromatic)

Cyclohexane (C_6H_{12})
(Chair conformation)

Aromatic

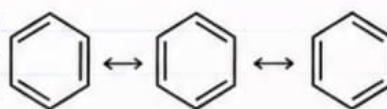
(Arenes)

Benzene (C_6H_6)
(Planar structure)

8. Aromatic Hydrocarbons (Arenes)

- Contain one or more benzene ring.
- Obey Hückel's rule $\rightarrow (4n + 2) \pi$ electrons ($n = 0, 1, 2, \dots$).
- Most stable due to resonance.
- General formula (monocyclic): C_nH_{2n-6} ($n \geq 6$)

Benzene - Structure

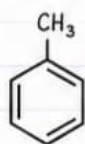


Kekulé structures (resonance)

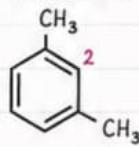
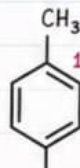
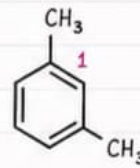
Planar ring
(120°)

9. Nomenclature of Aromatic Hydrocarbons (Arenes)

(i) Monosubstituted Benzene

Methyl benzene
(Toluene)

(ii) Disubstituted Benzene

o-Xylene
(1,2-dimethylbenzene)
(ortho)p-Xylene
(1,4-dimethylbenzene)
(para)m-Xylene
(1,3-dimethylbenzene)
(meta)

In disubstituted benzene, if two substituents are equivalent, numbering is done to give the lowest set of locants.

10. Physical Properties

(i) Alkanes

- Non-polar, colourless, odourless.
- Insoluble in water, soluble in organic solvents.
- Boiling point $\uparrow$ with $\uparrow$ in molecular mass.
- Physical state:

$C_1 - C_4$	Gases
$C_5 - C_{17}$	Liquids
C_{18} and above	Solids (waxy)

(ii) Alkenes & Alkynes

- Physical properties similar to alkanes.
- Slightly higher b.p. than alkanes of comparable molecular mass.
- Lower members are gases, higher members are liquids/solids.

(iii) Aromatic Hydrocarbons

- Have higher boiling points due to resonance and π -electron cloud.
- Usually colourless liquids or solids with characteristic odour.
- Insoluble in water, soluble in organic solvents.

11. Chemical Properties (Summary Table)

Type of Hydrocarbon	Combustion	Substitution	Addition	Oxidation	Polymerisation
Alkanes	Burn in excess O_2 to give CO_2 and H_2O .	Halogenation (Cl_2/Br_2 , UV light)	No (they are saturated)	Hard to oxidise; combustion only.	No
Alkenes	Burn to give CO_2 and H_2O .	Less reactive towards substitution.	Yes (H_2 , X_2 , HX , H_2O) at $C=C$ bond.	Easily oxidised ($KMnO_4$, etc.)	Yes (Addition polymerisation)
Alkynes	Burn to give CO_2 and H_2O .	Less reactive towards substitution.	Yes (more reactive than alkenes).	Easily oxidised.	Yes (Addition polymerisation)
Aromatic Hydrocarbons	Burn to give CO_2 and H_2O .	Electrophilic substitution (e.g., nitration, sulfonation, halogenation).	No (due to stability of benzene ring).	Resistant to oxidation; requires strong conditions.	No

★ Point

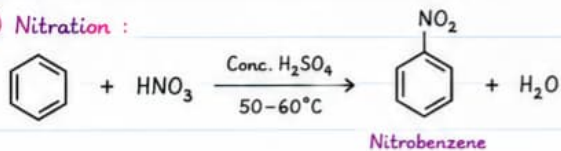
Stability order $\rightarrow$ Aromatic > Alkyne > Alkene > Alkane

★ Remember

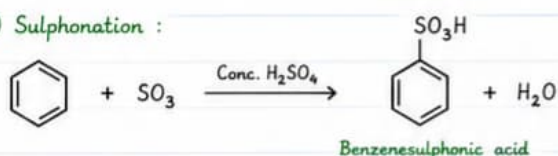
More C-C bonding $\rightarrow$ more stability.

12. Important Reactions of Aromatic Hydrocarbons (Arenes)

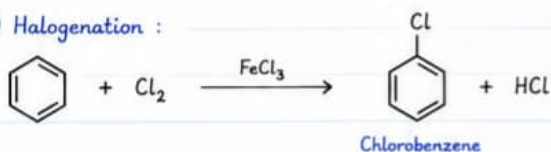
① Nitration :

Electrophile
in nitration
= NO_2^+

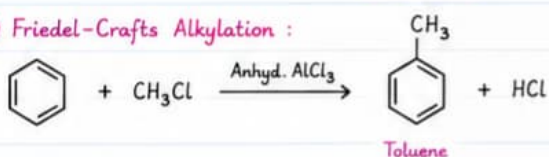
② Sulphonation :

Electrophile
in sulphonation
= SO_3

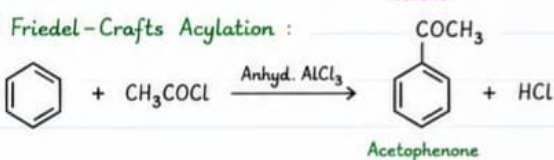
③ Halogenation :

Electrophile
in halogenation
= Cl^+

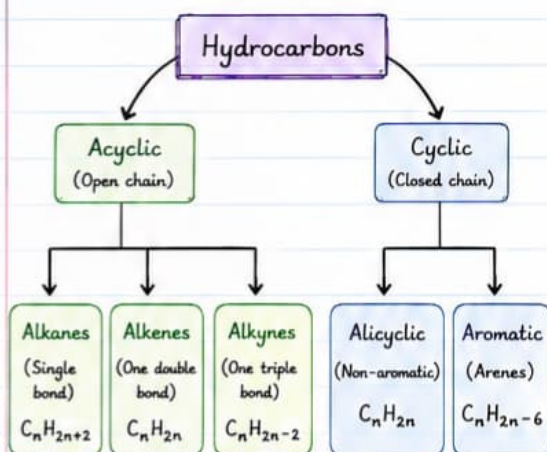
④ Friedel-Crafts Alkylation :

Electrophile
in alkylation
= R^+

⑤ Friedel-Crafts Acylation :

Electrophile
in acylation
= RCO^+

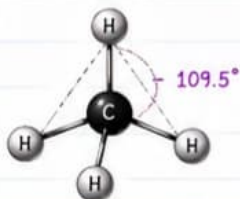
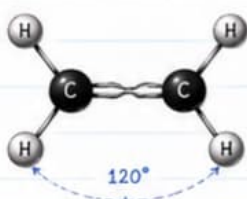
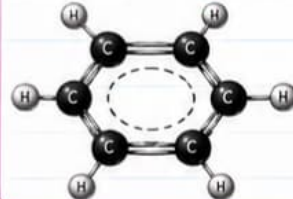
13. Summary Mind Map



Key Points

- Alkanes → Saturated, least reactive
- Alkenes → Unsaturated, more reactive
- Alkynes → Unsaturated, more reactive
- Aromatic → Resonance stabilized, shows electrophilic substitution.

14. 3D Structures

Methane (CH_4)
(Tetrahedral)Ethene (C_2H_4)
(Planar)Ethyne (C_2H_2)
(Linear)Benzene (C_6H_6)
(Planar ring)

15. Summary

- Hydrocarbons contain only C and H atoms.
- They are classified into **acyclic** and **cyclic**.
- Acyclic** → Alkanes (single bond), Alkenes (double bond), Alkynes (triple bond).
- Cyclic** → Alicyclic (non-aromatic), Aromatic (arenes).
- Alkanes** → Saturated, show substitution reactions.
- Alkenes & Alkynes** → Unsaturated, show addition reactions.
- Aromatic hydrocarbons** → Show electrophilic substitution reactions.
- Reactivity order : Alkynes > Alkenes > Alkanes.
- Stability order : Aromatic > Alkyne > Alkene > Alkane.

Quick Revision

- General formulae → Alkanes ($\text{C}_n\text{H}_{2n+2}$), Alkenes (C_nH_{2n}), Alkynes ($\text{C}_n\text{H}_{2n-2}$), Arenes ($\text{C}_n\text{H}_{2n-6}$)
- Alkanes → sp^3 , tetrahedral, 109.5°
- Alkenes → sp^2 , planar, 120°
- Alkynes → sp , linear, 180°
- Aromatic → Resonance, planar ring, 120°
- Electrophilic substitution in arenes.
- Addition reactions in alkenes & alkynes.

Keep
Practicing
& Stay
Consistent!

16. Important Points

- More the branching in alkanes, lower is the boiling point.
- Unsaturated hydrocarbons are more reactive than alkanes.
- Aromatic hydrocarbons are more stable due to resonance.
- In electrophilic substitution, ring remains intact.

Resonance in Benzene



17. Mnemonic

"ALL ALIENS ARE ALWAYS ACTIVE"

↓ ↓ ↓ ↓ ↓

Alkanes Alkenes Alkynes Alicyclic Aromatic

(Reactivity increases from left to right)